

Dhia Eddine Barhoumi

Computer Science Engineering Student at National Engineering School of Tunis (ENIT)

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PROFILE

Computer Science Engineering student at ENIT with hands-on experience in multi-agent AI, retrieval-augmented generation, computer vision, and infrastructure automation. Experienced in FastAPI, Spring Boot, and Angular, with a focus on modular architecture, validation of AI-generated code, and tested, containerized applications for scientific and cloud infrastructure workflows.

AREAS OF EXPERTISE

Multi-Agent AI, Retrieval-Augmented Generation, Computer Vision, Geospatial Analysis, Workflow Automation, Infrastructure as Code, Cloud & DevOps, Software Architecture

EDUCATION

- **National Engineering School of Tunis (ENIT)** 2023 – 2026
Engineering Degree in Software Engineering Tunis, Tunisia
- **Preparatory Institute for Engineering Studies of Monastir (IPEIM)** 2021 – 2023
Preparatory Cycle in Physics and Technology (PT) – High Honors (ranked 34/830) Monastir, Tunisia

INTERNSHIP EXPERIENCE

- **Direction Générale des Ressources en Eau (DGRE)** Summer 2026
Second-Year Engineering Internship – CartaGen Tunis, Tunisia
 - Developed CartaGen, a multi-agent platform for natural-language hydrological analysis, automated rainfall mapping, and rainfall yearbook generation for Tunisia’s DGRE.
 - Built a FastAPI application following hexagonal architecture, with supervisor-led agents for text-to-SQL queries, geospatial code generation, and reporting over PostgreSQL/PostGIS data.
 - Integrated semantic retrieval over station metadata and database schemas, cloud and local LLM providers, cached geospatial templates, and an automated code-correction loop.
 - Implemented spatial analysis using inverse distance weighting (IDW), Thiessen polygons, and rainfall anomaly calculations; automated map creation and national or regional yearbook generation in LaTeX/PDF.
 - Added AST-based validation of generated Python code, subprocess execution in per-request temporary directories, SQL input checks, and configurable CORS restrictions.
 - Containerized the application with Docker Compose and health checks; validated its workflows with 24 automated unit and integration tests.
- **Tunisie Télécom** Summer 2025
Full-Stack Developer Intern Kef, Tunisia
 - Engineered a web-based IT asset management platform for tracking hardware and software licenses.
 - Built the backend using Java 21, Spring Boot 3.5, and JPA; migrated the database from H2 to PostgreSQL.
 - Developed a responsive Angular 20 interface with Material Design and role-based access control (RBAC).

RESEARCH, ACADEMIC AND PERSONAL PROJECTS

- **Net2Terraform – Vision-to-Infrastructure Pipeline** ENIT, Tunisia
PFA 2 / Research Project – Supervised by Wafa Mefteh
 - Developed a vision-to-infrastructure pipeline as a second-year project (2025), extracting structured JSON/YAML topology data from network drawings and automating lab deployments with Terraform and Containerlab.
 - Co-developed the multimodal Net2Terraform platform described in the accompanying research manuscript, converting topology diagrams and natural-language architecture descriptions into Terraform configurations for AWS.
 - Combined YOLOv8 detection, OpenCV link inference, and Tesseract OCR to extract network components, connections, and labels into a shared architecture model.
 - Implemented hybrid retrieval using SentenceTransformers, FAISS, BM25, and cross-encoder reranking to ground conversational infrastructure generation in technical documentation.

- **Network Automation & Monitoring**

Personal Project

◦ Created Python and Jinja2-based automation scripts, using Ansible for large-scale configuration deployment across virtualized network infrastructures.

◦ Implemented post-deployment monitoring to verify network state compliance.

2025
- **SocialMedia – Microservices Architecture**

Personal Project

◦ Built a distributed system with seven independent services using Node.js, TypeScript, and MongoDB Atlas.

◦ Managed inter-service communication through a centralized API Gateway, emphasizing domain-driven design and horizontal scalability.

2025
- **RendezvousJEE – Clinic Management**

Personal Project

◦ Built a Jakarta EE 10 application using stateless EJB components, JPA, and Hibernate for persistence.

◦ Deployed the application on WildFly with Maven lifecycle management.

2025

TECHNICAL SKILLS

- Programming: Python, Java, TypeScript

• Backend: FastAPI, Pydantic, Spring Boot, Node.js, Jakarta EE, JPA/Hibernate, EJB

• Frontend: Angular, Material Design, HTML/CSS, JavaScript

• Databases: PostgreSQL/PostGIS, MongoDB Atlas, H2, SQL/NoSQL

• Architecture: Hexagonal architecture, microservices, REST APIs, domain-driven design

• Validation / Testing: AST analysis, SQL input checks, CORS, unit and integration tests

• Computer Vision: YOLOv8, OpenCV, Tesseract OCR, Roboflow

• AI / Agents: Supervisor orchestration, text-to-SQL, LLM code generation, Ollama

• Retrieval: RAG, SentenceTransformers, FAISS, BM25, reranking

• Geospatial / Reporting: IDW, Thiessen polygons, rainfall analysis, LaTeX/PDF

• Cloud / Automation: AWS, Terraform, Ansible, Jinja2, Containerlab

• Tools / DevOps: Docker Compose, Git, CI/CD, Maven, WildFly

RESEARCH MANUSCRIPT

- **Dhia Eddine Barhoumi**, Wael Hammali, and Wafa Mefteh.

From Network Diagrams to Deployable Cloud Infrastructure: A Multimodal AI-Based Approach.

Research manuscript, National Engineering School of Tunis, ICT Department, RISC Laboratory, University of Tunis El Manar.

Outcome: Paper accepted at IEEE AIDIST 2026 – "L'intelligence artificielle pour une transition numérique et durable", 5th FCIERI Forum (29–31 Oct. 2026, Tunis).

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- **CCNA: Introduction to Networks** – Cisco Networking Academy (ENIT)

June 23, 2025

• **Hashgraph Developer Course** – The Hashgraph Association (Hedera)

April 23, 2025

• **Efficient Large Language Model (LLM) Customization** – NVIDIA

Certificate of Competency

October 21, 2024